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Midterm Progress: M0–M2 Environment, Component-Control Audit, and Data Pipeline

Scope: milestones M0–M2. Revised: 2026-10-29. First written 2026-10-27, before the CUDA audit completed.

This is a progress report. No task-accuracy experiment has been run and there is no Δ result. Every statement is labelled as one of:

- Measured — produced by code that actually ran, cited to its source.
- Implemented, not run — code exists but has not executed on target hardware or real data.
- Estimated — planning figures with no evidential weight.
- Proposed — design choices to be checked in the pilot and frozen before the main run.

Related documents: the protocol is in EXPERIMENT_PROTOCOL.md, the milestone plan in IMPLEMENTATION_PLAN.md, and the proposal in PROPOSAL_SNAPSHOT.md. The full audit report is M1_MAC_FINDINGS.md, the pipeline handoff is M2_PIPELINE.md, the real-data validation is M2_REAL_DATA_VALIDATION.md, and the architecture and compute analysis is PROJECT_DEEP_DIVE.md.

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0. Headline for this reporting period

M1 is complete and accepted. The project's single largest scientific risk — whether the vision encoder, connector, and language backbone can be quantized independently, rather than a loader merely accepting a flag — has been resolved by a measured per-module audit of all eight configurations on an NVIDIA T4. Independent component control works. No model or quantizer fallback is needed.

M2 is implemented and validated on real POPE-COCO data. The corruption, split, and annotation pipeline exists, is tested, and has been checked against the official POPE files and CelebA annotations.

What is still absent: any accuracy number. No pilot has run, no real image has been through the model, and no severity, split size, or memory cap is frozen. The transfer-penalty measurement the project exists to produce has not started.

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1. Milestone status at a glance

Milestone	State
M0 — environment and access	Done. Mac development environment and Colab CUDA environment both pinned and measured

Milestone	State
M1 — component-control audit	Accepted. All 8 configurations audited on a T4; 331/331 linear modules converted as requested; savings exactly additive
M2 — data and corruption pipeline	Implemented and unit-tested (42 tests). POPE-COCO validated on real data; CelebA annotations validated, images not yet downloaded
M3 — pilot, cache, protocol freeze	Not started. Blocked only on team decisions listed in §7
M4 — selection and statistics	Not started
M5 — main runs and report	Not started

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2. Environment setup (M0)

The project needs two environments, because bitsandbytes NF4 requires an NVIDIA CUDA GPU and has no Apple Silicon backend.

	Development environment	Experiment environment
Host	Apple Silicon Mac	Google Colab, NVIDIA Tesla T4 (free tier)
Purpose	Write code, inspect modules, run unit tests, FP16 load on CPU	FP16 and NF4 GPU inference, memory measurement
Python	3.11.17 (Conda env `mpvs-m1`)	3.13.16
PyTorch	2.14.1	2.11.0+cu130 (CUDA 13.0)
Transformers	4.52.4	4.52.4 (matched)
Accelerate	1.15.0	1.15.0
bitsandbytes	not available	0.50.2
Compute capability	n/a (`mps_available=False`)	7.5
State	Measured: FP16 checkpoint loads on CPU; `pytest` clean	Measured: all 8 configurations built and ran (§4)

Pinned model

```
Model:     HuggingFaceTB/SmolVLM2-2.2B-Instruct
Revision:  482adb537c021c86670beed01cd58990d01e72e4  (model and processor)
```

Transformers 5.20.1 was tried and rejected after its SmolVLM image-token merger failed. 4.52.4 is the recorded working version on both hosts. This is a real compatibility finding, not a version preference: it was discovered by running the smoke test, not by reading release notes.

Reproducing

```
# development
conda env create -f environment-macos.yml
conda activate mpvs-m1
python -m pip install -e .
PYTHONPATH=src python -m pytest -q
python scripts/preflight.py --output artifacts/preflight-macos-m1.json
```

For the experiment environment, open `notebooks/m1_colab_cuda_nf4.ipynb` in Colab, select a T4 GPU, and run all cells. The notebook installs the project, runs FP16 / uniform-NF4 / language-NF4 as a feasibility gate, then runs all eight candidates in fresh subprocesses, preserves failure JSON, summarizes acceptance, and downloads the artifacts. Generated artifacts stay untracked.

2.1 First GPU measurement: FP16 only (measured, 2026-10-27)

This was the project's first GPU evidence, recorded before the full audit and retained because it independently reproduces the FP16 reference.

Measurement	Value
Load time (includes first download)	143.1 s
Peak allocated after load	4,504,592,896 B
Peak reserved during inference	5,322,571,776 B
Synthetic inference (first call, 4 new tokens)	3.92 s

Two cross-checks fall out of this, and both passed:

- Peak allocated after load is ~11 MB above the measured 4,493,569,888 bytes of FP16 weights — consistent with weights plus small runtime overhead.
- 5,322,571,776 bytes is 4.957 GiB, which is exactly the peak reserved figure the independent full audit later recorded for the FP16/FP16/FP16 candidate (§4). Two separately written runs agree to the byte.

The first attempt at this notebook failed at processor loading because `num2words` was missing from the Colab install list. An earlier attempt also cloned a published commit that predated the M1 implementation and stopped at `pip install -e .`. Both were infrastructure failures that produced no evidence; the clone cell now fails early with a stale-branch diagnostic. Recording them matters because the protocol distinguishes infrastructure failures from model answers.

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3. Pipeline flow

3.1 End to end

```
flowchart TD
    A["POPE-COCO + CelebA<br/>official annotations"] --> B["Normalize records<br/>image ID, ident
    B --> C["Split by original image<br/>and by CelebA identity"]
    C --> P["Pilot<br/>~100 questions"]
    C --> D["Development"]
    C --> T["Test<br/>held out"]
    D --> X["Corrupt each split separately<br/>clean + darkness x3 + blur x3"]
    T --> X
    M["Pinned SmolVLM2-2.2B<br/>FP16 checkpoint"] --> Q["Build 8 configs<br/>vision / connector /
    Q --> G{"Audit gate<br/>is requested precision<br/>actually realized?"}
    G -- "no" --> R["Report blocker<br/>or use disclosed subset"]
    G -- "yes" --> I["Inference, batch size 1<br/>fixed prompt, greedy decoding"]
    X --> I
    I --> K[("Resumable prediction cache")]
    K --> S["Select config per memory cap<br/>clean / target / 25-50-75% mixed<br/>DEVELOPMENT RO
    S --> E["Evaluate selected configs<br/>on the same target test images"]
    K --> E
    E --> F["Delta + paired cluster bootstrap<br/>+ selection stability"]
```

M0–M2 cover everything above the inference step. The audit gate has now been passed; M3–M5 (pilot freeze, caching, selection, statistics) come next.

3.2 The M1 audit gate

```
flowchart TD
    A["For each of the 8 requested configs"] --> B["Load fresh from original checkpoint<br/>never
    B --> C["Apply NF4 only to supported linear layers<br/>of the requested components"]
```

```
C --> D["Dump every module: class, stored dtype,<br/>quant_state, exceptions"]
D --> E{"Requested == realized?"}
E -- "no" --> F["Mark unsupported, record reason"]
E -- "yes" --> H["Measure unique weight bytes<br/>+ NF4 metadata"]
H --> I["One synthetic inference<br/>record peak allocated / reserved"]
I --> J{"Ran without failure?"}
J -- "no" --> F
J -- "yes" --> K["Compute effective signature<br/>merge duplicate configs"]
K --> L{"Some memory cap admits<br/>at least 2 distinct configs?"}
L -- "yes" --> M["M1 accepted"]
L -- "no" --> N["Scientific blocker: report it"]
```

Every branch of this gate was exercised. The outcome was the left path at every decision point.

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4. What M1 measured

All figures in this section are measured on a Tesla T4, each candidate loaded freshly from the same original checkpoint.

4.1 Component control is real

Component	Supported linear modules	Converted to NF4	Exceptions
Vision encoder (`model.vision_model`)	162	162	none
Connector (`model.connector`)	1	1	none
Language backbone (`model.text_model`)	168	168	`lm_head` kept FP16
Total	331	331	one documented output-head exception

Across all eight audit reports: every requested NF4 module is a real `bitsandbytes Linear4bit` converted from `torch.nn.Linear`; every one reports NF4 storage, FP16 compute, uint8 quantized storage, and no double quantization; every requested FP16 component has zero quantized modules; `lm_head` stayed FP16; every mismatch list is empty; no unclassified module was quantized; and all eight effective signatures are distinct.

This is the sentence the protocol demanded: the audit validates independent component control, and it is not global quantization relabelled as mixed precision.

4.2 Measured weight storage

Vision	Conn.	Backbone	Weight bytes	GiB	Peak reserved GiB	Gen. s
NF4	NF4	NF4	1,556,898,080	1.450	2.283	3.98
NF4	FP16	NF4	1,587,421,408	1.478	2.283	4.05
FP16	NF4	NF4	2,147,801,504	2.000	2.600	3.90
FP16	FP16	NF4	2,178,324,832	2.029	2.627	3.89
NF4	NF4	FP16	3,872,143,136	3.606	4.182	3.98
NF4	FP16	FP16	3,902,666,464	3.635	4.213	3.85
FP16	NF4	FP16	4,463,046,560	4.157	4.781	4.01

Vision	Conn.	Backbone	Weight bytes	GiB	Peak reserved GiB	Gen. s
FP16	FP16	FP16	4,493,569,888	4.185	4.957	3.78

All eight returned a successful deterministic generation on a synthetic 384×384 image. This checks execution viability only — it is not accuracy, robustness, real-image, or throughput evidence.

4.3 The strongest check: savings are exactly additive

Holding the other two components at FP16, moving one component to NF4 saved:

Component	Bytes saved
Vision encoder	590,903,424
Connector	30,523,328
Language backbone	2,315,245,056

`4,493,569,888 − 590,903,424 − 30,523,328 − 2,315,245,056 = 1,556,898,080`

That is exactly the measured all-NF4 total. Perfect additivity means the three component controls do not contaminate each other — independently confirming §4.1 at the storage level, by a different route.

4.4 Storage is fully accounted for

For the all-NF4 candidate:

Category	Bytes
Parameter storage	1,429,194,464
Persistent buffers	128
Exposed quantization metadata	127,703,488
Total	1,556,898,080

127,703,488 bytes across 2,042.9 M quantized parameters is 0.0625 bytes/param — exactly one FP32 `absmax` per 64-value block, consistent with double quantization being off. Nothing in the storage total is unexplained.

4.5 The prediction held

The architecture analysis derived these totals from the published `config.json` before any GPU was available. Every candidate landed within 1.4 MB of its measured value, a maximum error of 0.090%. The storage model is therefore validated, and the memory cost of future candidate changes can be computed without spending GPU time.

4.6 Usable memory caps

Cap (bytes)	Eligible candidates	Vision varies	Backbone varies
1,587,421,408	2	no	no
2,147,801,504	3	yes	no
2,178,324,832	4	yes	no
3,872,143,136	5	yes	yes
3,902,666,464	6	yes	yes

M1 acceptance required only one cap admitting two distinct candidates, and 1,587,421,408 bytes satisfies it. But it is the weakest scientific choice: its two candidates differ only in connector precision, which is 0.9% of the weights. The caps admitting four and six candidates vary vision and backbone precision and are the useful comparisons. Proposed, not frozen.

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5. M2 — data and corruption pipeline

M2 needs no GPU and proceeded in parallel with M1.

5.1 Order of operations (this order is the leakage guarantee)

1. Build normalized records: `dataset, original_image_id, identity_id, question_id, question, label`.
2. Split first, by whole group, into pilot / development / test with a fixed seed. Save manifests with checksums.
3. Then generate corrupted variants. Every variant inherits its original image's split.
4. Never let a group appear in two splits, even if split sizes end up uneven.

Grouping is correct by construction: the split group resolves to the CelebA identity and to the POPE original image, so no identity and no image can straddle a split.

5.2 POPE-COCO: validated on real data

The three official POPE COCO files and the 500 referenced COCO val2014 images were downloaded and checked. Measured, for every sampling variant:

Check	Result
Questions	3,000
Distinct images	500
Yes/no balance	1,500 / 1,500
Questions per image	6
Duplicate question IDs	none
Missing images	none
Loader defaults vs. schema	matched, no code change needed
Normalization	deterministic

5.3 CelebA: annotations validated, images pending

Attribute and identity files match their published MD5 values. Measured: 202,599 images, 40 attributes, 10,177 identities, and every image has an identity.

615 identities carry inconsistent `Male` labels across their images. This is dataset label noise to document, not repair — silently correcting source labels would be an undisclosed change to the benchmark.

The images themselves are not yet downloaded, so no CelebA split is final.

5.4 Validation results

Trial splits on POPE adversarial and CelebA Eyeglasses (placeholder seed): zero leakage, byte-identical on rerun. Trial POPE pilot corruptions: 420 variants, completeness valid, 0 pixel mismatches across two renders.

Two real defects were found by validation review and fixed:

1. `validate_variant_inheritance` reported `valid: true` when some clean/darkness/blur variants were missing, which silently breaks paired evaluation. It now reports missing variants, inconsistent corruption specs, and undeclared specs, and the CLI passes declared specs so a condition dropped for every question is also caught.

2. Splits did not detect the same image bytes appearing under two image IDs or two identities in different splits. Groups sharing a source-image checksum are now merged into one indivisible unit, and the validator reports checksum leakage.

Nine synthetic-fixture tests were added; eight of them fail against the previous code. The second defect is the dangerous class — a leakage hole that raises no error and would have invalidated the study silently.

A third defect was found in the corruption specification and fixed: `CorruptionSpec.value` was not normalized, so `2` and `2.0` produced different signature digests and `True` was accepted as a blur radius (`bool` subclasses `int`). One transform could hold three distinct cache keys. Since cache keys must include the image transformation, this was a correctness defect. `value` is now coerced to `float`, booleans are rejected, and two regression tests cover it.

Command: `PYTHONPATH=src python3 -m pytest -q` → 42 passed (Python 3.13, macOS, CPU). Synthetic fixtures only; synthetic success is not research evidence.

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6. Risks, with current status

#	Risk	Status	Mitigation
1	GPU access	Resolved. Free-tier Colab T4 ran the full audit	Keep the Mac for code and tests; T4 is the known-working path
2	Component control may not work	Resolved. 331/331 modules, zero mismatches, additive savings	Keep the audit as a gate whenever the grid or library versions change
3	Throughput	Open — now the top risk. Only measurement is 3.78–4.05 s/answer, a cold first call including warm-up	At 4 s, ~95,000 answers is ~106 GPU-hours; at 1 s, ~26. Measure steady state in the pilot before committing
4	Blur severity is not comparable across domains	Open — newly identified. Gaussian radius is in absolute pixels; a CelebA crop is ~178 px wide and a COCO image ~640 px, so one radius is far stronger on faces	Define radius relative to the short edge, or declare severities per-domain and never compare a raw severity across domains
5	Double quantization shifts every cap	Open decision. Currently off, costing ~128 MB of metadata	Freeze before the pilot alongside the cap definitions
6	Colab session limits	Open	Batch size 1, one config per process, resumable cache so a disconnect loses at most one item. At the pessimistic throughput, free tier is not viable — budget a paid runtime

#	Risk	Status	Mitigation
7	Severity window	Open	The pilot must show accuracy declining with severity without collapsing to chance; choose levels on pilot data only
8	Connector result is ambiguous	Open by construction	The connector is a single 21 M-param layer, so a null cannot distinguish “no effect” from “lever too small”. Report the weaker reading
9	Selection noise	Open	Report selection frequencies under development bootstrap alongside Δ
10	Invalid answers bias the result	Open	Count invalid as incorrect, freeze the parser before selection, report invalid rate per configuration
11	8 GB assumption	Largely resolved	Largest configuration peaked at 4.957 GiB with one synthetic image; re-check with real images in the pilot
12	Dataset access and terms	Open	CelebA images stay internal under its research-use agreement; release code and split identifiers only
13	Schedule	Open	The proposal’s Oct 12 protocol freeze and Oct 19 main-run dates have passed. M1 and M2 are now done, so the critical path is the pilot

7. Next actions

Decisions first — none of them need a GPU, and the pilot cannot be frozen without them:

Action	Needs GPU
Freeze NF4 double quantization on or off	No
Decide blur severity units: absolute pixels or relative to image size	No
Choose the POPE sampling variant and the CelebA attribute set	No
Approve the identity-capped balanced CelebA sampling rule, split sizes, and seed	No
Download CelebA images under the research-use agreement; build final manifests	No

Action	Needs GPU
Run the human answerability review on corrupted pilot variants	No
Measure steady-state throughput, then choose severities and caps	Yes
Implement the resumable prediction cache before any scaled run	Yes

Nothing above should be reported as GPU-tested or as an experimental result until the corresponding run has completed and its outputs are recorded.

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8. Honest summary

The infrastructure phase of this project is finished and it is in better shape than the schedule suggests. Component control — the assumption the whole study rests on — is not assumed but measured, at the level of all 331 individual linear modules, with an independent additivity cross-check and an exact match to a prediction made beforehand. The data pipeline is implemented, and validation review has already caught three defects, one of which was a silent leakage hole.

What remains is the experiment itself, and it has not started. There is no accuracy figure, no Δ, and nothing frozen. The honest position is that the project is roughly two weeks behind the proposal's timeline on results while being ahead of where most projects are on verification, and that the single number which decides whether the full design is affordable — steady-state inference throughput — is still unmeasured.